LMA Assignment 1

Baseline Retrieval vs. Cross-Encoder Re-Ranking

Doshik Rahul Repaka (2026701035)

Abstract

This report evaluates whether Cross-Encoder re-ranking improves evidence selection in a retrieval-augmented generation (RAG) pipeline. The experiment uses six manually specified questions over a cleaned, chunked copy of the Constitution of India. Both systems share the same Bi-Encoder Top-10 candidate set; the treatment is the ordering of those candidates before selecting the three passages supplied to the language model. Mean reciprocal rank (MRR@10) increases from 0.2625 to 0.8667 and Top-3 evidence hit rate increases from 33.3% to 83.3%. However, one relevant passage is demoted from rank 2 to rank 5, and median retrieval latency rises from 0.797 s to 35.336 s. The results support a conditional conclusion: reranking substantially improves evidence placement in this small test set, but it is neither uniformly beneficial nor latency efficient in the measured execution environment.

1 Research question and experimental contrast

The central question is not whether a Cross-Encoder produces higher scores, but whether it places the passage needed to answer a query inside the language model's limited context window. The two pipelines are:

1. **Baseline:** encode the query, retrieve the ten highest cosine-similarity candidates, and pass the first three to the generator.
2. **Re-ranked:** retrieve the same ten candidates, score each query–passage pair with a Cross-Encoder, sort by that score, and pass the first three to the same generator.

This is a controlled comparison of second-stage ordering, not a comparison of different candidate-generation systems. Since the candidate set is shared, re-ranking can improve the final context only when the needed passage is present in the initial Top 10. It cannot repair a first-stage recall failure.

2 Corpus and implementation

The selected book for this experiment is *Constitution of India*, no OCR or something is done here only PDF processing is applied for it. It holds 402 pages after cleaning following were removed:

- 331 repeated header/footer lines
- 623 page-number lines
- 281 decorative lines

The final resultant was 406 chunks, where target size is 500 words with 100 overlap.

The notebook reports 1,354.59 s for embedding/index preparation and 2.97 s for vector upload. These are setup costs, not per-query retrieval latency, and should not be conflated with the online timing comparison.

Table 1: Recorded system configuration

Component	Configuration recorded in notebook
Bi-Encoder	`mixedbread-ai/mxbai-embed-large-v1`
Cross-Encoder	`mixedbread-ai/mxbai-rerank-base-v2`
Embedding matrix	406×1024
Vector index	Qdrant Cloud; cosine distance; 406 chunks
Generator	Gemini API, configured as `gemini-3.5-flash-lite`
Candidate/context sizes	Top 10 candidates; Top 3 passages to generator

3 Mechanism and hypotheses

A Bi-Encoder independently maps queries and passages into vectors. Passage vectors can be computed once and searched efficiently using cosine similarity. A Cross-Encoder processes the query and passage jointly, allowing direct token-level interactions that may distinguish close semantic matches, exceptions, and constraints more effectively. Its cost is higher because each candidate pair must be evaluated for a query.

In Bi-encoder architecture it always tries to map queries and passages into vectors. Once the passage vectors are computed once and then they are searched efficiently. A cross-encoder processes the query and passage jointly, allowing direct token-level interactions that may distinguish close semantic matches, exceptions and constraints more effectively but this comes with a higher cost.

The experiment tests three practical expectations:

- **H1:** re-ranking increases the fraction of queries whose gold evidence is in the Top 3.
- **H2:** gains are most visible when the relevant passage is initially ranked 4–10.
- **H3:** these gains have a measurable latency cost and may include regressions.

4 Ranking results

Here all the ranking results act differently because recall@10 only checks the first hit things due to that we can't distinguish the things well. In case of mrr@10 it sees the possible answers which are ground-truth relevant things but issue is that it doesn't depend on the position. Where as the nDCG@10 measures the top-k list based on how well it was ranked against it's ideal ranking and it's position is taken into consideration.

Table 2: Aggregate ranking metrics over six queries

Metric	Baseline	Re-ranked	Absolute change
Recall@10	1.0000	1.0000	0.0000
MRR@10	0.2625	0.8667	+0.6042
nDCG@10	0.4367	0.8978	+0.4611
Top-3 hit rate	33.3%	83.3%	+50.0 percentage points

The unchanged Recall@10 is expected: both systems re-order the same Top-10 candidates. It is therefore not evidence that re-ranking has no effect; it shows that the measured improvement occurs after candidate retrieval. MRR rises by approximately 230% relative to baseline, while Top-3 hit rate improves from two of six queries to five of six. These are large changes in this sample, but

six hand-selected questions do not provide a reliable estimate of performance on an unseen query distribution.

Table 3: Gold evidence rank changes and Top-3 consequences

ID	Failure/query type	Gold chunk (page)	Rank shift	Interpretation
Q1	Keyword trap: dissolution of Council of States	`chunk_00069` (70)	$8 \rightarrow 1$	Miss in baseline Top 3; recovered after re-ranking.
Q2	Exception: ordinance when both Houses are in session	`chunk_00086` (87)	$5 \rightarrow 1$	Exception clause enters generator context.
Q3	Constraint: nominated MPs voting in Presidential election	`chunk_00056` (57)	$4 \rightarrow 1$	Elected-member restriction is surfaced.
Q4	Buried evidence: non-member appointed Minister	`chunk_00065` (66)	$6 \rightarrow 1$	Six-month qualification becomes available.
Q5	Regression: retired Supreme Court judge acting in court	`chunk_00088` (89)	$2 \rightarrow 5$	Evidence is lost from reranked Top 3.
Q6	Double prosecution/punishment	`chunk_00041` (42)	$3 \rightarrow 1$	Evidence remains in Top 3 and improves position.

Q1–Q4 are the strongest evidence for the proposed mechanism: each gold passage begins outside the baseline context and is promoted to rank 1. Q6 is a positive control, not a baseline failure: its evidence was already in the baseline Top 3 and remained there. Q5 is the counterexample. The gold passage starts at rank 2 but is demoted to rank 5, changing a correct evidence selection into a Top-3 miss. Thus, the Cross-Encoder is not simply a monotonic correction layer; its scores can reorder a relevant passage below less useful candidates.

5 Answer-level analysis

Q1: Council of States dissolution. The baseline response says the supplied context is insufficient. The re-ranked response answers that the Council of States is not subject to dissolution and cites page 70. This is an evidence-availability improvement rather than proof that the generator itself became more capable: the decisive passage was absent from baseline Top 3 and present in re-ranked Top 3.

Q2: Ordinance exception. The baseline response abstains for insufficient context. The re-ranked response gives the exception that an ordinance cannot be promulgated when both Houses are in session, citing page 87. This is a consequential retrieval error type: a general description of ordinance power is not enough when the query asks about an explicit exception.

Q3: Nominated members and Presidential election. The baseline response reports that the supplied context does not establish the voting rule. The re-ranked context surfaces Article 54's use of elected members, allowing the answer to distinguish elected from nominated members. The gain depends on retrieving the precise qualifier, not merely passages mentioning Parliament or elections.

Q4: Minister who is not a member of either House. The re-ranked answer gives a qualified "yes" and states that the Minister ceases to hold office after six consecutive months without membership. This qualification is essential: reporting only that a non-member can be appointed would omit the condition that makes the answer accurate.

Q5: Retired Supreme Court judge. The baseline answer gives the prohibition and cites the relevant constitutional restriction. After re-ranking, the gold passage falls to rank 5 and the generator reports insufficient context. This is a direct end-to-end regression and demonstrates why mean metrics should be accompanied by query-level error analysis.

Q6: Double prosecution and punishment. Both answer contexts contain the relevant Article 20(2) passage, and both answers state the prohibition. Re-ranking improves the passage from rank 3 to rank 1, but it does not change the substantive answer. This shows that a rank improvement is not necessarily an answer-quality improvement when the baseline already supplies adequate evidence.

Across the cases, the evidence supports a narrow causal interpretation: changes in answer sufficiency align with whether the relevant chunk enters the Top 3. It does not establish that the Cross-Encoder eliminates hallucinations. The tested baseline often abstains rather than fabricating a rule, and the sample is too small to quantify hallucination rates.

6 Latency and cost of the second stage

Table 4: Median per-query retrieval latency

Measurement	Median
Baseline retrieval	796.8 ms
Cross-Encoder stage	34,572.7 ms
Total re-ranked retrieval	35,335.6 ms
Additional overhead	34,525.2 ms
Overhead relative to baseline	4,332.8%

- Reranking takes approximately 44.3× longer than baseline retrieval, with most added latency coming from the Cross-Encoder.
- Timings exclude LLM generation and model-loading costs and reflect only the recorded environment. Limited hardware and runtime details prevent reliable generalization to other machines.
- Reranking improves Top-3 hit rate by 50 percentage points, increasing the likelihood of retrieving relevant evidence for the LLM.
- However, the additional median latency of approximately 34.5 seconds may be unsuitable for interactive applications.
- Further investigation should focus on Cross-Encoder execution, device placement, batching, and timing methodology. These are recommendations; their effectiveness has not been tested in this experiment.

7 Why not use a Cross-Encoder over the full collection?

- Two-stage retrieval: The Bi-Encoder efficiently retrieves the Top 10 candidates, and the Cross-Encoder reranks them to improve the ordering of relevant passages.
- Limitation: The Cross-Encoder cannot recover relevant passages missing from the initial Top 10. Candidate recall and ranking quality must therefore be evaluated separately.

8 Validity threats and limits

- **Small, selected sample:** six manually constructed questions are adequate for illustrating specific failure modes, not for estimating broad generalization.
- **Single gold chunk per query:** a query may be answerable from multiple passages, but the metric treats the designated chunk as the relevant target.
- **Text extraction quality:** the first extracted page contains visibly garbled characters in the Hindi title. Although 402 pages yielded text, extraction noise could affect indexing and retrieval; legal clauses should be checked against the source.
- **Reranker regression:** Q5 shows that aggregate gains conceal a case where relevant evidence is demoted.
- **Latency reproducibility:** the measured Cross-Encoder time is high relative to baseline; hardware, device selection, warm-up, and timing boundaries should be documented before operational conclusions are drawn.
- **Generator dependence:** answer comparisons involve both retrieved context and a stochastic/API-based generator. They do not isolate every possible source of answer variation.

9 Conclusion

For the six tested questions, Cross-Encoder re-ranking substantially improves the ordering of manually identified evidence: MRR@10 increases from 0.2625 to 0.8667, nDCG@10 from 0.4367 to 0.8978, and Top-3 hit rate from 33.3% to 83.3%. The most meaningful gains occur in Q1–Q4, where the relevant passage is promoted from ranks 4–8 to rank 1 and the generator moves from insufficient context to a supported answer. However, Q5 demonstrates that re-ranking can remove relevant evidence from the final context, and the observed latency penalty is substantial.

The defensible conclusion is therefore conditional: re-ranking improved evidence selection on this small, targeted evaluation, but the experiment does not prove universal answer improvement or hallucination elimination. A deployment decision would need a larger query set, explicit evidence annotations, regression monitoring, and reproducible latency measurements.

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